

Hemoglobin in the Trenches

A century of attempts to put hemoglobin into a wounded body, from a postpartum ward in 1949 to an Army laboratory in the 1980s, and the mistakes that each generation made in its own way

The history of oxygen carriers is usually told as a sequence of products: this one failed, that one was withdrawn, another went bankrupt. That telling is accurate and misses what is most useful about the past, which is the sequence of ideas. Each generation of researchers believed it understood why its predecessors had failed. Each was partly right and partly wrong, and the wrong part turned out to be the one that mattered. This essay follows the ideas rather than the companies, and it pays attention to the moments at which a mistaken explanation held a field in place for a decade.

I will be careful throughout about what the sources establish. The history here draws on a narrative review of hemoglobin-based oxygen carriers and on accounts of the Army's research program, and where dates or numbers matter I give them as those sources do.

The first thought: hemoglobin is the oxygen, so give hemoglobin

The idea appears early, and it is easy to see why. Hemoglobin is the molecule that carries oxygen in blood, and red cells are the awkward package that holds it. If a patient is bleeding and blood is unavailable, why not extract the hemoglobin and inject it?

The first recorded attempts date to the end of the nineteenth century. A German physician, von Stark, gave hemoglobin under the skin to anemic patients in 1898 and could not obtain a stable preparation. In the years before and during the First World War, researchers refined the preparations. A pair of American physicians, Sellards and Minot, infused hemoglobin solutions into thirty-three subjects, in volumes of a few milliliters up to about thirty. Every recipient passed hemoglobin in the urine, a visible sign that the kidney was clearing it. Yet thirty reported no overt harm. It was an encouraging result for the wrong reason: the doses were tiny, and the kidney's reaction was mistaken for tolerance.

The first idea was thus simple and incomplete. It treated hemoglobin as a fuel and neglected the vessel, the organ and the immune system through which it would pass.

The case that should have taught more

In 1949 a group led by Amberson published one of the most sobering cases in the whole literature. A woman had suffered a postpartum hemorrhage, compatible blood could not be found, and the physicians gave her a stroma-free hemoglobin solution, hemoglobin purified from red cells with the membrane fragments, the stroma, removed. The review I am following describes what happened: the patient improved initially, then developed oliguria, a sharp fall in urine output, and died of renal failure.

It is a haunting account because it holds both halves of the story. The hemoglobin worked: it raised her oxygen-carrying capacity while nothing else was available. It also destroyed her

kidneys. The same observation would be made again, in animals and in people, for the next fifty years, and the explanation would change three times.

The second idea: it is the stroma

After the war, researchers noticed that the injury to the kidney seemed worse when the hemoglobin preparation was crude. The obvious suspect was the debris from broken red cells. Fragments of membrane are known to activate clotting and cause damage to small vessels, and a solution containing them could plausibly injure the kidney.

The idea produced a program of purification. Researchers in the 1960s, including Rabiner, showed in dogs that a well-purified, stroma-free hemoglobin could serve as an oxygen-carrying replacement in hemorrhagic shock and could sustain animals through exchange transfusions. The review describes his results as showing the solution to be lifesaving in dogs undergoing exchange transfusion. It looked as though the problem had been identified and solved.

Then came the human data. In the late 1970s, studies by Savitsky and colleagues gave purified stroma-free hemoglobin to healthy volunteers. The review is direct about the results: all developed oliguria with reduced creatinine clearance; two reported abdominal pain; seven developed a slow heart rate and high blood pressure. The kidney was affected even though the stroma were gone. The second idea was wrong, or at least incomplete.

The third idea: it is the dimer

The next explanation came from chemistry. A hemoglobin molecule outside a red cell is not stable as a four-chain unit. In dilute solution it splits into two-chain halves, called dimers, and the dimers are small enough to pass through the filter of the kidney. Once in the tubules they can precipitate, and they were held responsible for the kidney injury.

That explanation led directly to the chemical modifications that defined the next generation. If the tetramer could be prevented from splitting, the dimers would never form. Chemists therefore developed ways to cross-link the four chains together, and to polymerize several tetramers into larger assemblies. The diaspirin cross-link, which bridges the two alpha chains, was one of these. Glutaraldehyde polymerization, used in the bovine products, was another.

The third idea was much closer to the truth, and it produced the products that reached the clinic. But it too had an incomplete part. Cross-linking prevented the kidney injury from dimers, and yet the new products still raised blood pressure. The hypertension would need a fourth explanation.

The Army laboratory

At this point in the history a particular institution enters. The United States Army had a strong interest in a blood substitute that could be carried to a battlefield, stored without refrigeration, and given without typing. It supported research at the Letterman Army Institute of Research and elsewhere, and its scientists helped develop the diaspirin cross-linked hemoglobin. The review describes that program as a collaborative effort between the pharmaceutical company Baxter and the Army, and says that Baxter later continued alone.

A published account from the Letterman program bears a title that summarizes the state of the field as it stood: a review of modified hemoglobin research at Letterman, attempts to delineate the toxicity of cell-free tetrameric hemoglobin. The word delineate is honest. The researchers were trying to separate the effects of the molecule into parts, to say what came from the kidney, what from the vessels and what from contamination. They were doing what a good research program does, and it is instructive that after so many years they still framed the task as delineating toxicity.

The fourth idea: it is the nitric oxide

The explanation for hypertension arrived in the 1990s and was, in retrospect, elegant. The lining of blood vessels makes nitric oxide, a gas that relaxes the smooth muscle in the vessel wall and keeps the vessel open. Free hemoglobin binds nitric oxide with extraordinary avidity, several orders of magnitude tighter than oxygen binds. When hemoglobin is loose in plasma, it can slip through gaps in the vessel lining, capture the nitric oxide near its source, and remove it. Without the gas, the vessels constrict, the pressure rises and blood flow to organs falls.

This account explained the hypertension, the reduced flow to some organs, the abdominal pain from constriction of the gut vessels, and much else. It also suggested a design principle: make the hemoglobin too big to leave the vessel lumen, and it will not reach the source of the gas. The bovine glutamers and the PEGylated products can be read as attempts to apply that principle.

Yet the fourth idea also turned out to be an incomplete story. Even large molecules seemed to produce some vasoconstriction, and it became clear that the gas is present in the lumen and not just in the wall. The hypothesis remains a leading explanation, and it is discussed elsewhere in this series. What I want to note here is the pattern: each generation's insight was real, and each was mistaken in believing it complete.

What the human trials showed

Meanwhile, human trials proceeded, and their results were disturbing. The review reports the outcome of the trauma trial of the diaspirin product published in 1999: within twenty-eight days, twenty-four of the fifty-two patients who received the product had died, forty-six percent, against eight of forty-six, seventeen percent, who received saline. A trial of a polymerized human hemoglobin in trauma, published years later, reported thirty-day mortality of thirteen percent in the treated group against ten percent in the control group, a difference that did not reach conventional significance but pointed in an unfavorable direction. A pooled analysis in 2008, combining sixteen trials, reported increased risks of death and of heart attack.

These results have been analyzed to exhaustion. I want only to observe how they interacted with the ideas. Each product had been designed on the basis of the explanation then current. Each failed for reasons that the explanation had not anticipated. The pattern suggests that the theory was always a step behind the biology.

The fifth idea, still forming

The field today holds a set of hypotheses rather than a single explanation. Nitric oxide scavenging is one. Oxidation of the iron, which produces free heme and reactive species, is another. The interaction of free hemoglobin with the endothelium and with platelets, the effect

on the immune system, the role of oxygen affinity in triggering the arteriolar constriction described elsewhere on this site, the assay interference that clouds measurement, and the heterogeneity of the polymer mixtures are all in play.

What distinguishes the present from the past is the tools. Researchers can now measure nitric oxide directly in tissue, they can characterize polymer distributions precisely, they can image microvessels, and they can compare molecules head to head in standardized animal models. None of these guarantees success, but they raise the odds that the next explanation will be tested before it is bet on.

The lessons of the sequence

I draw four lessons from the history, and I offer them as instruction to anyone entering the field now.

Distrust the complete explanation. Every theory in the sequence was partial. The moment an explanation is presented as sufficient, it stops being examined, and the unexplained residue becomes the next disaster.

Human data corrects animal data. The dog results of the 1960s were sound and did not predict the human response. Species differ in kidney handling, in vascular reactivity and in immune response. A compound that behaves well in animals has passed the first exam, not the last.

Small doses mislead. The early human studies used doses that were too small to show harm and gave a false comfort. Toxicity that appears at high doses was not visible at low doses, and the doses that matter clinically are the high ones.

Institutions matter. The Army's interest was a driver of the research, and its willingness to pursue the problem over decades produced knowledge that a commercial firm alone might not have generated. It also shows that a strong sponsor is not a guarantee of success. Both statements are true.

A note on the war that started it

The title of this essay refers to trenches because the impulse behind the field has always come from war. Sellards and Minot worked in the era of the First World War, when the question of what to give a soldier who had lost blood at the front was pressing. The Army's laboratory programs of the late twentieth century were driven by the same question. The current Department of Defense interest in prehospital trials is its descendant.

That lineage carries a moral. The people who wanted a battlefield oxygen carrier were not chasing a curiosity. They were trying to keep young people alive who otherwise would die. A century of failures has not reduced the strength of that motive, and it explains why the effort has continued through so many discouraging results. The history should discourage credulity and should not discourage persistence.

What the past asks of the present

If the pattern holds, the next generation of researchers will discover that today's explanations were incomplete. The question is how to be ready for that. I would suggest three practices drawn from the sequence above.

Design studies that can detect harm early, with adequate sample sizes, independent monitoring and pre-specified stopping rules.

Keep the mechanisms open. Do not treat any single explanation as the answer, and measure more than one candidate cause in every study.

Preserve the record. Much of the knowledge in this essay comes from a handful of review articles, and it is fragile. A patient community that supports an archive of the field's history, with the original reports accessible, protects the next generation from repeating the last one.

Why the early doses misled

It is worth pausing on the arithmetic of caution. A few milliliters of hemoglobin solution is a trivial fraction of what a hemorrhaging adult needs. The blood volume of an adult is roughly five liters, and a resuscitation that hopes to replace a meaningful share of lost oxygen capacity must deliver quantities that are orders of magnitude larger than the early experiments used. When the dose rises by a factor of a hundred, effects that were invisible at the low end can become dominant. The kidney can clear a small load without difficulty and may be overwhelmed by a large one. A vessel that shrugs off a trace of nitric oxide scavenging may clamp down when the scavenging is sustained.

This is a general feature of pharmacology, and it is easy to forget when a field is young and eager. The lesson for readers of the historical record is to ask, of any early safety claim, what dose it applied to and what dose the eventual use would require. A finding of no harm at a tenth of the intended dose is a finding about the tenth.

The problem of names

A second difficulty in reading this history is terminological. The same substance has been called stroma-free hemoglobin, cell-free hemoglobin, modified hemoglobin, hemoglobin solution, blood substitute and oxygen therapeutic, and the field has argued at length over which label is honest. A blood substitute implies that the product does everything blood does, which none of them do. An oxygen carrier implies a narrower and more accurate claim. In this series the term is BHOC when the source is bovine and the discussion is of the class, and the older names are used only where they belong to a particular historical report.

The naming matters because names shape expectations. Patients, physicians and regulators who hear substitute expect a replacement and are disappointed by a bridge. Those who hear carrier expect a limited tool and judge it accordingly. Much of the disillusionment recorded in the literature came from expectations set too high by the vocabulary.

What the archives make easy to miss

Reading the review literature, one is struck by how much of the history depends on a few sentences repeated from paper to paper. The account of the 1949 case is short. The account of the volunteer studies is shorter still. It is entirely plausible that details have been lost or simplified in retelling, and a careful reader should treat every secondhand summary, including the ones in this essay, as an invitation to consult the original.

There is also a survivorship problem in the record. Programs that ended quietly, with a few disappointing animal experiments and no publication, left almost no trace. What we know is weighted toward the programs that reached humans or generated controversy. The true history includes many unpublished dead ends, and the reader should assume that the ideas visible in the literature are only the ones that were pursued far enough to be written down.

The patient inside the history

It is easy, in a narrative of ideas, to forget that each entry in the record was a person. The woman in 1949 had a baby and a hemorrhage and a physician who reached for the only thing available. The volunteers in the 1970s agreed to an experiment and lay in a laboratory while their urine output fell. The trauma patients in the 1990s were enrolled under an exception to informed consent, because they were too injured to agree, and the community consultation that accompanied such trials is a subject of its own.

Keeping those people in view changes the tone of the analysis. It becomes harder to speak of a failed generation as if it were a set of errors on a chart. The investigators were, on the whole, serious people trying to save lives, and the participants gave something real. The best way to honor both is to extract from the record everything it can teach, and to say so plainly.

A short chronology, for orientation

For readers who want the sequence in one place, here is the outline, as the sources give it. In 1898 hemoglobin was first tried in anemic patients. In the era of the First World War, Sellards and Minot infused small volumes into a group of subjects. In 1949 the postpartum case ended in renal failure. In the 1960s dogs survived exchange transfusion with purified hemoglobin. In the late 1970s human volunteers developed oliguria with the stroma-free preparation. Through the 1980s and 1990s, chemical modification produced cross-linked and polymerized products, with Army and industry collaboration. In 1999 the diaspirin trauma trial reported its mortality difference. In 2008 the pooled analysis appeared. Bovine products meanwhile reached veterinary use and limited human approval in two countries. Each of these dates is drawn from the reviews cited below, and the reader is encouraged to verify them.

Further reading

The history of hemoglobin-based oxygen carriers is reviewed in a narrative article published in *Annals of Blood*, on which the dates and case details above are based. The Army program is described in a report titled *Review of Modified Hemoglobin Research at Letterman*, published in *Artificial Cells, Blood Substitutes, and Biotechnology*. The 1999 trauma trial of the diaspirin cross-linked product appeared in the *Journal of the American Medical Association*, and the

pooled analysis in the same journal in 2008. Figures quoted are from the review and should be checked against the original reports.